Precession of Particle Orbits: A Point Mass Embedded in an Extended Mass Distribution

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A test particle orbiting a fixed point mass M_p has the equation of motion $\ddot{\boldsymbol{r}} = -GM_p\boldsymbol{r}/r^3$. Now suppose that a perturbing acceleration, $\boldsymbol{P}$, is applied to the particle, such that the modified equation of motion is

$$\ddot{\boldsymbol{r}} + \frac{GM_p}{r^3}\boldsymbol{r} = \boldsymbol{P} \ . \tag{1}$$

We suppose here that the perturbation is small, i.e., that $r^2|\boldsymbol{P}|/GM_p \ll 1$. In the special case where $\boldsymbol{P}$ is due to a smooth, spherically symmetric distribution of mass centered on M_p, we have

$$\boldsymbol{P} = -\frac{GM_e(r)}{r^3}\boldsymbol{r} \ , \tag{2}$$

where $M_e(r)$ is the mass in the extended component within a radius r. In the pure Kepler problem ($\boldsymbol{P} = 0$), there are two conserved vector quantities:

$$\boldsymbol{h} = \boldsymbol{r} \times \boldsymbol{v} \ , \tag{3}$$

and

$$\boldsymbol{e} = \frac{\boldsymbol{v} \times \boldsymbol{h}}{GM_p} - \frac{\boldsymbol{r}}{r} \ , \tag{4}$$

where $\boldsymbol{h}$ is the orbital angular momentum, and $\boldsymbol{e}$ is the Laplace-Runge-Lenz (LRL) vector, which points in the direction of pericenter and has a magnitude equal to the eccentricity. Since $\boldsymbol{P}$ in eq. (2) is central, $\boldsymbol{h}$ remains a constant of the motion for $\boldsymbol{P} \neq 0$. However, $\boldsymbol{e}$ is no longer conserved. For a general central perturbation, it is straightforward to show that the time derivative of $\boldsymbol{e}$ is given by

$$GM_p\dot{\boldsymbol{e}} = \boldsymbol{P} \times \boldsymbol{h} \ . \tag{5}$$

A fairly accurate picture of the perturbed motion is that of a precessing ellipse with fixed semimajor axis, a, and eccentricity, e. The orbit-averaged change in $\boldsymbol{e}$ reflects the secular precession of the perturbed elliptical motion. The orbit-averaged value of some function f can be expressed as

$$\langle f \rangle = \frac{1}{T}\int_0^T dt f \simeq \frac{1}{Th}\int_0^{2\pi} d\phi r^2 f \ , \tag{6}$$

where T is the period of the bounded radial orbit, and we have used $r^2\dot{\phi} = h$, where ϕ is the polar angle (true anomaly) in the plane of motion. The second integral above is only approximate, since the orbit does not close after a full azimuthal cycle. Hereafter, we will assume that $|\boldsymbol{P}|$ is sufficiently small that the ϕ integral in eq. (6) is a suitable orbit average, and that T and h take their Keplerian values:

$$T = 2\pi\left(\frac{a^3}{GM_p}\right)^{1/2} \ , \tag{7}$$

and

$$h = [GM_p a(1 - e^2)]^{1/2} \ . \tag{8}$$

For a vector $\boldsymbol{A}$ of fixed length that rotates with constant angular velocity $\boldsymbol{\omega}$, the equation of motion is $\dot{\boldsymbol{A}} = \boldsymbol{\omega}\times\boldsymbol{A}$. Therefore, the angular precession rate of the perturbed orbit is defined by the relation

$$\langle \dot{\boldsymbol{e}} \rangle = \boldsymbol{\omega} \times \boldsymbol{e} \ . \tag{9}$$

If $\boldsymbol{\omega}\cdot\boldsymbol{h} > 0$ we say that the precession is *prograde*, while if $\boldsymbol{\omega}\cdot\boldsymbol{h} < 0$ the precession is *retrograde*. Since $\boldsymbol{h}$ is strictly constant for a central perturbation, we find that $GM_p\langle\dot{\boldsymbol{e}}\rangle = \langle\boldsymbol{P}\rangle\times\boldsymbol{h}$.

With these generalities in hand, we can proceed by specifying the functional form of $\boldsymbol{P}(r)$. Suppose that the extended mass distribution is characterized by a power-law density profile, $\rho = \rho_0(r/r_0)^{-\gamma}$, so that $M_e(r) = M_0(r/r_0)^{3-\gamma}$ for $\gamma < 3$, where $M_0 = M_e(r_0)$. Now the orbit-averaged perturbing acceleration is

$$\langle\boldsymbol{P}\rangle = -\frac{GM_0}{r_0^{3-\gamma}}\left\langle\frac{\boldsymbol{r}}{r^\gamma}\right\rangle \ . \tag{10}$$

The integral over ϕ in eq. (6) is computed assuming the unperturbed elliptical motion. A natural coordinate system for the Kepler problem is one in which $\boldsymbol{e}$, $\boldsymbol{h}\times\boldsymbol{e}$, and $\boldsymbol{h}$ constitute an orthogonal set of basis vectors. We denote the corresponding orthonormal basis vectors by $\hat{\boldsymbol{x}}\equiv\hat{\boldsymbol{e}}$, $\hat{\boldsymbol{y}}\equiv\hat{\boldsymbol{h}}\times\hat{\boldsymbol{e}}$, and $\hat{\boldsymbol{z}}\equiv\hat{\boldsymbol{h}}$, where hats indicate unit vectors. In this system of units, the position vector is simply $\boldsymbol{r} = r(\cos\phi\hat{\boldsymbol{x}} + \sin\phi\hat{\boldsymbol{y}})$, where $r = l/(1+e\cos\phi)$ and $l = a(1-e^2)$. In the unperturbed Kepler problem, the vectors $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$ are fixed. In the perturbed problem, take $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$ to be constant vectors that point in the directions of $\boldsymbol{e}$ and $\boldsymbol{h}\times\boldsymbol{e}$ at time $t = 0$. The term proportional to $\hat{\boldsymbol{y}}$ clearly vanishes in the orbit average, so that we are left with

$$\langle\boldsymbol{P}\rangle = -\hat{\boldsymbol{x}}\frac{GM_e(l)}{Th}\int_0^{2\pi} d\phi\frac{\cos\phi}{(1+e\cos\phi)^{3-\gamma}} \ . \tag{11}$$

Examination of eqs. (5) and (9) reveals that

$$\boldsymbol{\omega} = \frac{1}{T}\frac{M_e(l)}{M_p}\frac{I(\gamma,e)}{e}\hat{\boldsymbol{z}} \ , \tag{12}$$

where $I(\gamma, e)$ is the integral in eq. (11). The angle through which the orbit precesses in a single period is then

$$\Delta\omega = \frac{M_e(l)}{M_p}\frac{I(\gamma,e)}{e} \ , \tag{13}$$

where the precession is prograde or retrograde if $I(\gamma, e)$ is positive or negative, respectively.

A specific and relevant example is the case where $\gamma = 2$ (e.g., a singular isothermal sphere). The ϕ integral is analytic, with the result

$$\frac{I(2,e)}{e} = -\frac{2\pi}{e^2}\left[\frac{1}{(1-e^2)^{1/2}} - 1\right] < 0 \ . \tag{14}$$

For this choice of γ, the precession is retrograde. For arbitrary γ and $e \ll 1$, we find that

$$\frac{I(\gamma, e\ll 1)}{e} = -\pi(3-\gamma), \tag{15}$$

and so we generally expect retrograde precession for $\gamma < 3$. Figure 1 shows $-I(\gamma,e)/[\pi e(3-\gamma)]$ as a function of e for 11 different values of γ, from $\gamma = 0.25$ (top curve) to $\gamma = 2.75$ (bottom curve). We see from Fig. 1 that the right-hand side of eq. (15) is within a factor of order unity of the correct value of $I(\gamma,e)/e$ for $e \lesssim 0.6$.

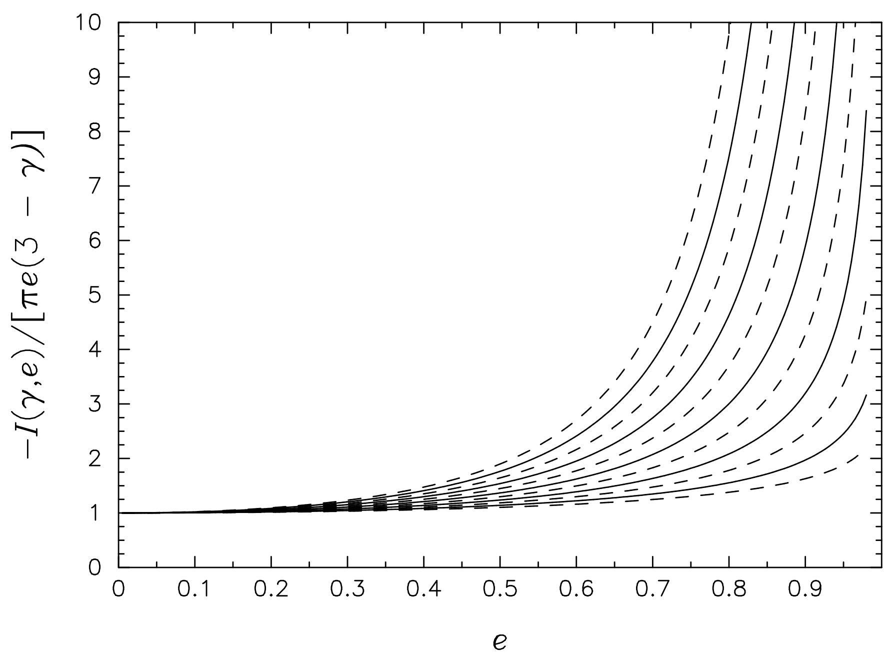

Figure 1: The 11 curves correspond to a range of $\gamma = 0.25$ (top curve) to $\gamma = 2.75$ (bottom curve), in steps of 0.25.